A finite-history convergence certificate for Chambolle–Pock at $\theta = 1/8$

Abstract

For the literal primal-first Chambolle–Pock iteration with extrapolation parameter 1/8, we prove boundedness, square summability of first differences, and weak convergence to a saddle point whenever $\tau\sigma\|L\|^2 \le 148537/50000 = 2.97074$. The result holds on arbitrary real Hilbert spaces for proper lower-semicontinuous convex functions, a nonempty KKT set, and arbitrary initial points. A finite-history storage inequality, certified by 126 explicit rational coefficients, supplies both actual-history coercivity and strict descent. Its constant-history scalar quadratic is positive definite. The convergence proof uses only its positive diagonal entries and the vanishing coupling pairing on differences of KKT points to separate weak cluster points. All coefficients and exact positive-definiteness pivots are printed; an independent original-map checker accompanies the source. The bound is sufficient, not asserted sharp, and leaves a gap to the known three-cycle obstruction.

Keywords: Chambolle–Pock iteration; weak convergence; finite-history storage; exact rational certificate; Hilbert space; near-critical step size

2020 Mathematics Subject Classification. Primary 65K10; Secondary 90C25, 47J25.

1 Statement and scope

Let $\mathcal{H}, \mathcal{G}$ be real Hilbert spaces, $L : \mathcal{H} \to \mathcal{G}$ bounded linear, and f, h proper lower-semicontinuous convex extended-real-valued functions. For $a > 0$, the proximal mapping is

$$\operatorname{prox}_{af}(z) = \arg\min_x \{f(x) + \|x - z\|^2/(2a)\}.$$

It is defined everywhere and has a unique value. Put

$$\mathcal{Z} = \{(x, y) : -L^*y \in \partial f(x),\ Lx \in \partial h(y)\}.$$

For $h = g^*$ this is the usual primal–dual saddle set. The two updates in this report are exactly

$$\begin{aligned} x_n &= \operatorname{prox}_{\tau f}(x_{n-1} - \tau L^* y_{n-1}), \\ y_n &= \operatorname{prox}_{\sigma h}\big(y_{n-1} + \tfrac{\sigma}{8} L(9x_n - x_{n-1})\big), \qquad n \ge 1. \end{aligned} \tag{1}$$

No separately initialized extrapolated variable is used.

Theorem 1. *Assume $\mathcal{Z} \ne \varnothing$, $\tau, \sigma > 0$, and*

$$\tau\sigma\|L\|^2 \le R := \frac{148537}{50000}.$$

For every $(x_0, y_0) \in \mathcal{H} \times \mathcal{G}$, the sequence generated by (1) is bounded,

$$\sum_{n=1}^{\infty} \bigl(\|x_n - x_{n-1}\|^2 + \|y_n - y_{n-1}\|^2 \bigr) < \infty,$$

and (x_n, y_n) converges weakly to an element of $\mathcal{Z}$. In finite dimensions convergence is in norm.

Initial values $f(x_0), h(y_0)$ may be infinite. There is no smoothness, strong convexity, uniqueness, finite-dimensionality, or preliminary boundedness assumption. The theorem includes $L = 0$. It makes no norm-convergence assertion in general Hilbert space.

2 Raw proximal inequalities, before storage

Keep the fixed certificate parameter R, even when the actual product is smaller. Define

$$X_n = \sqrt{R/\tau}\, x_n, \quad Y_n = y_n/\sqrt{\sigma}, \quad K = \sqrt{\tau\sigma/R}\, L, \tag{2}$$

so $\|K\| \le 1$. Write $F(X) = f(\sqrt{\tau/R}\, X)$ and $H(Y) = h(\sqrt{\sigma} Y)$. Direct optimality of the two proximal subproblems gives

$$\begin{aligned} (X_{n-1} - X_n)/R - K^* Y_{n-1} &\in \partial F(X_n), \\ Y_{n-1} - Y_n + K(9X_n - X_{n-1})/8 &\in \partial H(Y_n). \end{aligned} \tag{3}$$

Fix any normalized KKT point $p = (U, V)$; thus $-K^*V \in \partial F(U)$ and $KU \in \partial H(V)$. Put $u_n = X_n - U$, $v_n = Y_n - V$ and define the centered gaps

$$\begin{aligned} F_p(u) &= F(U + u) - F(U) + \langle K^* V, u \rangle, \\ H_p(v) &= H(V + v) - H(V) - \langle KU, v \rangle. \end{aligned}$$

Both are nonnegative, vanish at zero, and have zero subgradient at zero. At every positive index their values are finite, and (3) gives the actual subgradients

$$\begin{aligned} a_i &= (u_{i-1} - u_i)/R - K^* v_{i-1} \in \partial F_p(u_i), \\ b_i &= v_{i-1} - v_i + K(9u_i - u_{i-1})/8 \in \partial H_p(v_i). \end{aligned} \tag{4}$$

Consequently, for positive i, j,

$$\begin{aligned} F_p(u_j) - F_p(u_i) - \langle a_i, u_j - u_i \rangle &\ge 0, \\ H_p(v_j) - H_p(v_i) - \langle b_i, v_j - v_i \rangle &\ge 0, \\ F_p(u_i) \ge 0, \quad \langle a_i, u_i \rangle - F_p(u_i) &\ge 0, \\ H_p(v_i) \ge 0, \quad \langle b_i, v_i \rangle - H_p(v_i) &\ge 0. \end{aligned} \tag{5}$$

These inequalities, derived from the original map, are the only nonlinear information used by the certificate. In particular, local history label zero below does not mean the possibly infeasible initial point.

3 Exact formal certificate

For this section only, let ξ_i, η_i $(0 \le i \le 4)$ be arbitrary formal vectors in one real Hilbert space, with scalar value symbols φ_i, ψ_i. Interleave the slots:

$$z = (\xi_0, \eta_0, \dots, \xi_4, \eta_4), \qquad \phi = (\varphi_0, \psi_0, \dots, \varphi_4, \psi_4).$$

Define, for $1 \le i \le 4$,

$$A_i = (\xi_{i-1} - \xi_i)/R - \eta_{i-1}, \qquad B_i = \eta_{i-1} - \eta_i + (9\xi_i - \xi_{i-1})/8.$$

The anchor $*$ has zero point, value, and gradient. Set

$$\begin{aligned} s(i, j, 0) &= \varphi_j - \varphi_i - \langle A_i, \xi_j - \xi_i \rangle, \\ s(i, j, 1) &= \psi_j - \psi_i - \langle B_i, \eta_j - \eta_i \rangle. \end{aligned}$$

When $i = *$ these mean the target value. A target $j = *$ uses zero point and value. No gradient at local index zero is used.

For $\ell = 3, 4$ define the ordered list $\mathcal{S}_\ell$ as follows: first take type $t = 0$, then type $t = 1$; for each type list

$$(*, j, t)\ (j = 0, \dots, \ell), \qquad (i, *, t)\ (i = 1, \dots, \ell),$$

then (i, j, t) in increasing $i = 1, \dots, \ell$ and, within each i, increasing $j = 0, \dots, \ell$ with $j \ne i$. Thus $|\mathcal{S}_4| = 50$ and $|\mathcal{S}_3| = 32$.

Appendix A specifies a symmetric 8×8 matrix P, a vector $c \in \mathbb{R}^8$, and positive multipliers $\lambda \in \mathbb{R}^{50}$ and $\mu \in \mathbb{R}^{32}$. Let

$$\begin{aligned} E_{\text{old}} &= \sum_{a,b=0}^{7} P_{ab} \langle z_a, z_b \rangle + \sum_{a=0}^{7} c_a \phi_a, \\ E_{\text{new}} &= \sum_{a,b=0}^{7} P_{ab} \langle z_{a+2}, z_{b+2} \rangle + \sum_{a=0}^{7} c_a \phi_{a+2}. \end{aligned}$$

The entries of c are signed. Positivity of P is neither claimed nor needed.

Lemma 1. *There are symmetric rational* 10×10 *matrices* D, C *such that, identically in all vector and value slots,*

$$\begin{aligned} E_{\text{old}} - E_{\text{new}} &= \sum_{k=1}^{50} \lambda_k s(\mathcal{S}_{4,k}) + \langle z, Dz \rangle, \\ E_{\text{old}} - \|\xi_3\|^2 - \|\eta_3\|^2 &= \sum_{k=1}^{32} \mu_k s(\mathcal{S}_{3,k}) + \langle z, Cz \rangle. \end{aligned} \tag{6}$$

They obey $C \succeq 0$ *and* $D \succeq \varepsilon Q$, *where* $\varepsilon = 10^{-6}$ *and* $\langle z, Qz \rangle = \|\xi_4 - \xi_3\|^2 + \|\eta_4 - \eta_3\|^2$.

Proof. Here are complete reconstruction and positivity tests, so the identities do not depend on unspecified matrices. Let e_a be the standard coordinate columns in $\mathbb{R}^{10}$, $e_{*,t} = 0$, and $e_{i,t} = e_{2i+t}$. For a label (i, j, t) put $d = e_{j,t} - e_{i,t}$. If $i = *$ set $S = 0$. Otherwise put

$$g_{i,0} = (e_{i-1,0} - e_{i,0})/R - e_{i-1,1}, \quad g_{i,1} = e_{i-1,1} - e_{i,1} + (9e_{i,0} - e_{i-1,0})/8,$$

and $S = -(g_{i,t}d^\top + dg_{i,t}^\top)/2$. The slack is exactly $\langle z, Sz\rangle + d^\top\phi$. Embed P in coordinates 0:7 to obtain P_o, and in 2:9 to obtain P_n. Embed c analogously as c_o, c_n. Define

$$D = P_o - P_n - \sum_k \lambda_k S_{4,k},$$
$$C = P_o - e_6e_6^\top - e_7e_7^\top - \sum_k \mu_k S_{3,k}.$$

Exact substitution of Appendix A gives

$$c_o - c_n = \sum_k \lambda_k d_{4,k}, \qquad c_o = \sum_k \mu_k d_{3,k},$$

which proves every value-coefficient cancellation in (6). All multipliers are positive, with minima

$$\min\lambda = \frac{7382473}{100000000}, \qquad \min\mu = \frac{71886295299}{100000000}.$$

Let Z stack five copies of I_2, and set

$$T = \begin{pmatrix} & & & I_8 \\ -I_2 & -I_2 & -I_2 & -I_2 \end{pmatrix}.$$

Exact substitution also gives $DZ = QZ = 0$, $T^\top Z = 0$, $\operatorname{rank} T = 8$, and zero last two rows and columns of C. The eight successive Schur pivots of $T^\top(D - \varepsilon Q)T$ and of $C_{0:7,0:7}$ are the strictly positive rationals in Appendix B. Successive elimination therefore proves both matrices positive definite. Since the columns of T span $\ker Z^\top$ and both D, Q vanish on $\operatorname{ran} Z$, the full-space inequalities follow. The supplied checker reconstructs every coefficient and pivot using exact rational arithmetic. Scalar matrix positivity applies equally to Hilbert-valued slots: expand the finite family of vectors in an orthonormal basis of its finite-dimensional span and sum the scalar inequalities. ∎

4 Contraction transfer and actual-history coercivity

A formal gradient above need not itself be a subgradient of an embedded function. Only its pairings are transferred. In $\mathcal{E} = \mathcal{G} \oplus \mathcal{H}$ define

$$Ju = (Ku, (I - K^*K)^{1/2}u), \qquad Iv = (v, 0).$$

The nonnegative square root exists because $\|K\| \le 1$. These are isometries and

$$\langle Ju, Iv\rangle = \langle Ku, v\rangle.$$

For $n \ge 4$ substitute

$$\xi_i = Ju_{n-3+i}, \quad \eta_i = Iv_{n-3+i}, \quad \varphi_i = F_p(u_{n-3+i}), \quad \psi_i = H_p(v_{n-3+i}).$$

Every same-type pairing is preserved by the isometries; every cross-type pairing is preserved by the displayed identity. Thus the formal slacks are exactly the genuine nonnegative quantities in (5). This covers every contraction, without invertibility or nonzero-norm assumptions on K.

Write $w_n = (Ju_{n-3}, Iv_{n-3}, \dots, Ju_n, Iv_n)$ and let ϕ_n list the corresponding eight actual gaps. The storage is the finite real number

$$\mathcal{E}_n(p) = \langle w_n, Pw_n\rangle + c^\top \phi_n.$$

All history indices are positive for $n \ge 4$, so no initial function values occur. Lemma 1 implies

$$\begin{aligned} \mathcal{E}_n(p) &\ge \|u_n\|^2 + \|v_n\|^2, \\ \mathcal{E}_n(p) - \mathcal{E}_{n+1}(p) &\ge \varepsilon\big(\|u_{n+1} - u_n\|^2 + \|v_{n+1} - v_n\|^2\big). \end{aligned} \tag{7}$$

In particular $\mathcal{E}_n(p)$ decreases to a finite nonnegative limit. The first inequality proves boundedness; telescoping the second proves square summability of all sufficiently late first differences. The finite prefix consists of finite-norm vectors and can be added. Hence

$$\Delta X_n := X_n - X_{n-1} \longrightarrow 0, \qquad \Delta Y_n := Y_n - Y_{n-1} \longrightarrow 0$$

in norm. This proves coercivity only on actual histories using their convex inequalities, not on arbitrary formal histories.

5 Gap decay and history collapse

The anchor inequalities from (5), followed by (4), give

$$\begin{aligned} 0 &\le F_p(u_n) + H_p(v_n) \le \langle a_n, u_n\rangle + \langle b_n, v_n\rangle \\ &= -\langle \Delta u_n, u_n\rangle/R - \langle \Delta v_n, v_n\rangle + \langle Ku_n, \Delta v_n\rangle + \tfrac{1}{8}\langle K\Delta u_n, v_n\rangle \longrightarrow 0. \end{aligned}$$

Boundedness and norm decay of the increments justify the limit. Thus both nonnegative gaps vanish individually. Every finite signed combination of their fixed-lag values tends to zero, including $c^\top \phi_n$.

Finite-lag differences tend to zero and the histories are bounded. Replacing every historical pair by the current pair therefore changes the quadratic storage by $o(1)$. Set $z_n = (X_n, Y_n)$ and let Z_4 stack four copies of I_2. Direct exact summation gives

$$M = Z_4^\top P Z_4 = \begin{pmatrix} 10648186303/2970740000 & -11266244157/2000000000 \\ -11266244157/2000000000 & 10648186303/1000000000 \end{pmatrix}. \tag{8}$$

Define on $\mathcal{H} \times \mathcal{G}$

$$q(a,b) = M_{11}\|a\|^2 + 2M_{12}\langle Ka, b\rangle + M_{22}\|b\|^2.$$

Then, for every normalized KKT point p,

$$\mathcal{E}_n(p) = q(z_n - p) + o(1), \qquad \lim_n q(z_n - p) \text{ exists and is finite.} \tag{9}$$

The scalar matrix M is positive definite: its first diagonal entry and determinant are positive, with

$$\det M = \frac{38232317384925803697194 87}{59414800000000000000000} > 0.$$

In fact, contraction of K and Cauchy–Schwarz give $q(a,b) \geq \lambda_{\min}(M)(\|a\|^2+\|b\|^2)$. Indeed one may bound the cross term below by $-2|M_{12}|\|a\|\|b\|$; the corresponding scalar matrix has the same eigenvalues as M. The cluster-point argument below only needs positivity on KKT differences, rather than this stronger global bound.

6 Weak cluster points and their uniqueness

Define the set-valued operator

$$\mathcal{T}(X,Y) = (\partial F(X) + K^*Y,\ \partial H(Y) - KX).$$

The product subdifferential is maximal monotone. Its bounded everywhere-defined skew linear perturbation is also maximal monotone; this is the standard full-domain continuous monotone perturbation theorem. From the original inclusions (3),

$$\rho_n = (-\Delta X_n/R + K^*\Delta Y_n,\ -\Delta Y_n + K\Delta X_n/8) \in \mathcal{T}z_n, \qquad \rho_n \to 0 \tag{10}$$

in norm. If $z_{n_j} \rightharpoonup z$, monotonicity with every $(a,b) \in \operatorname{gra}\mathcal{T}$ gives

$$\langle z_{n_j} - a, \rho_{n_j} - b\rangle \geq 0.$$

Since the sequence is bounded, weak convergence and strong residual decay allow passage to the limit. Thus $\langle z - a, -b\rangle \geq 0$ for every graph point, and maximality yields $0 \in \mathcal{T}z$. Every weak cluster point is therefore a normalized KKT point. Bounded Hilbert-space sequences have weakly convergent subsequences, so such a point exists.

Let $p = (U,V)$ and $p' = (U',V')$ be two KKT points. Monotonicity of ∂F and ∂H respectively yields

$$-\langle K(U-U'), V-V'\rangle \geq 0, \qquad \langle K(U-U'), V-V'\rangle \geq 0.$$

Hence the cross pairing is zero. Since the two diagonal entries of (8) are strictly positive,

$$q(p-p') = M_{11}\|U-U'\|^2 + M_{22}\|V-V'\|^2 > 0 \quad \text{if } p \neq p'. \tag{11}$$

This establishes the positivity on KKT differences needed below.

Let $\mathcal{Q}$ denote the bounded selfadjoint operator representing q; explicitly its blocks are $M_{11}I$, $M_{12}K^*$, $M_{12}K$, $M_{22}I$. If p, p' are weak cluster points, they are KKT and (9) shows that

$$q(z_n - p) - q(z_n - p') = -2\langle z_n, \mathcal{Q}(p - p')\rangle + q(p) - q(p')$$

has a limit. This expression is affine weakly continuous in z_n. Along subsequences converging weakly to p and p' its limits are respectively $-q(p - p')$ and $q(p - p')$. They must agree. Equation (11) forces $p = p'$.

Finally, a bounded Hilbert-space sequence with a unique weak cluster point converges weakly to it: otherwise some continuous linear functional separates a subsequence from that point by a fixed positive amount; a further weakly convergent subsequence contradicts uniqueness. This proves normalized full weak convergence. The invertible scalings (2) preserve boundedness, square summability, weak convergence, and KKT membership. Weak and norm convergence coincide in finite dimensions. This completes the proof of Theorem 1.

7 What the certificate does and does not establish

The certificate is fixed at $R = 2.97074$; smaller actual products are handled by contraction transfer, rather than by recomputing coefficients. It is a single sufficient bound for the whole indicated product interval. The present theorem is not derived from exclusion of cycles.

The previously established three-cycle obstruction occurs at

$$r_3 = \frac{88 + 16\sqrt{91}}{81}, \qquad R < r_3.$$

The strict comparison can be checked without decimal roots: $81R - 88 > 0$ and $(81R - 88)^2 < 256 \cdot 91$. The remaining numerical gap is $r_3 - R \approx 4.10156 \times 10^{-6}$. The interval (R, r_3) is not settled here; no globally sharp convergence threshold, convergence at r_3, or novelty claim is asserted. Neither exclusion of short cycles nor feasibility of a numerical search substitutes for the finite rational inequalities proved above.

A All 126 rational coefficients

Indices of P and c are zero-based; multiplier indices are one-based. The JSON lists the 36 upper-triangular entries of P row by row, then eight entries of c, then the 50 descent and 32 coercivity multipliers. The tables below print this entire data set, including every signed value coefficient. All omitted lower-triangular entries of P are obtained by symmetry. Labels $*$ and $t = 0, 1$ have exactly the meaning specified in Section 3.

Storage matrix, rows 0–2

Entry	Exact rational value
P_{00}	587893169959877769/29707400000000
P_{01}	−292084925571197/8000000000
P_{02}	7417570729990267873/148537000000000
P_{03}	3918277215907/31250000
P_{04}	−436279097537687153/5941480000000
P_{05}	208682181528223/2000000000
P_{06}	550074432989222107/148537000000000
P_{07}	354935414951/15625000
P_{11}	15308881094447/125000000
P_{12}	−2037813140613129/8000000000
P_{13}	82605136991893/1000000000
P_{14}	1474253253594357/16000000000
P_{15}	−435942433053339/2000000000
P_{16}	−717892541889689/16000000000
P_{17}	25795409652289/2000000000
P_{22}	−44630209551617/500000000
P_{23}	32045559813097/200000000
P_{24}	96720775694737/1000000000
P_{25}	−7305851983051/500000000
P_{26}	−1793655932451/31250000
P_{27}	101964586239137/1000000000

Storage matrix, rows 3–7

Entry	Exact rational value
P_{33}	$-19194297922707/1000000000$
P_{34}	$-102309612731521/500000000$
P_{35}	$2766500548131/50000000$
P_{36}	$-3458568922737/250000000$
P_{37}	$-23747636786149/200000000$
P_{44}	$-24661657875087/1000000000$
P_{45}	$4106001743209/200000000$
P_{46}	$54845645583/40000000$
P_{47}	$47569570229839/1000000000$
P_{55}	$229833514439281/1000000000$
P_{56}	$-1189347151727/40000000$
P_{57}	$-6718965069769/100000000$
P_{66}	$26161730465123/500000000$
P_{67}	$-37987913948909/500000000$
P_{77}	$173032779663047/1000000000$

Signed value coefficients

Entry	Exact rational value
c_0	$17274695259899/40000000$
c_1	$487928369681559/1000000000$
c_2	$-1428805202209/100000000$
c_3	$-134320452782577/1000000000$
c_4	$-22188711143139/250000000$
c_5	$-40263652135391/250000000$
c_6	$-164411205410507/500000000$
c_7	$-385105259963/2000000$

Descent multipliers, 1–25

Entry		Exact rational value
λ_{1}	$s(*,0,0)$	17151674157/4000000000
λ_{2}	$s(*,1,0)$	17056416943/4000000000
λ_{3}	$s(*,2,0)$	851098263/200000000
λ_{4}	$s(*,3,0)$	4248350893/1000000000
λ_{5}	$s(*,4,0)$	4240507623/1000000000
λ_{6}	$s(1,*,0)$	1335743367/250000000
λ_{7}	$s(2,*,0)$	5321495647/1000000000
λ_{8}	$s(3,*,0)$	5317463369/1000000000
λ_{9}	$s(4,*,0)$	2657220061/500000000
λ_{10}	$s(1,0,0)$	296832267/4000000000
λ_{11}	$s(1,2,0)$	702676010690167/1000000000
λ_{12}	$s(1,3,0)$	289877/3906250
λ_{13}	$s(1,4,0)$	15935866090343/200000000
λ_{14}	$s(2,0,0)$	12434269800831/62500000
λ_{15}	$s(2,1,0)$	37104257/500000000
λ_{16}	$s(2,3,0)$	78769389095893/100000000
λ_{17}	$s(2,4,0)$	74208513/1000000000
λ_{18}	$s(3,0,0)$	232914628349073/1000000000
λ_{19}	$s(3,1,0)$	81815839083227/500000000
λ_{20}	$s(3,2,0)$	18552129/250000000
λ_{21}	$s(3,4,0)$	6312141557879/10000000
λ_{22}	$s(4,0,0)$	148417/2000000
λ_{23}	$s(4,1,0)$	4314234563329/25000000
λ_{24}	$s(4,2,0)$	52375136186109/250000000
λ_{25}	$s(4,3,0)$	74208519/1000000000

Descent multipliers, 26–50

Entry		Exact rational value
λ_{26}	$s(*,0,1)$	4285072029/1000000000
λ_{27}	$s(*,1,1)$	4264255547/1000000000
λ_{28}	$s(*,2,1)$	4257505103/1000000000
λ_{29}	$s(*,3,1)$	4249814471/1000000000
λ_{30}	$s(*,4,1)$	264982841/62500000
λ_{31}	$s(1,*,1)$	83563967/15625000
λ_{32}	$s(2,*,1)$	2666101061/500000000
λ_{33}	$s(3,*,1)$	5316355083/1000000000
λ_{34}	$s(4,*,1)$	5299721513/1000000000
λ_{35}	$s(1,0,1)$	18456187/250000000
λ_{36}	$s(1,2,1)$	101409213201581/125000000
λ_{37}	$s(1,3,1)$	14764949/200000000
λ_{38}	$s(1,4,1)$	4079250226693/40000000
λ_{39}	$s(2,0,1)$	43238564420351/250000000
λ_{40}	$s(2,1,1)$	73824749/1000000000
λ_{41}	$s(2,3,1)$	2894893430271/3906250
λ_{42}	$s(2,4,1)$	14764949/200000000
λ_{43}	$s(3,0,1)$	39371209909831/125000000
λ_{44}	$s(3,1,1)$	50101378296073/500000000
λ_{45}	$s(3,2,1)$	73824747/1000000000
λ_{46}	$s(3,4,1)$	357417311002653/1000000000
λ_{47}	$s(4,0,1)$	7382473/100000000
λ_{48}	$s(4,1,1)$	23850567485847/125000000
λ_{49}	$s(4,2,1)$	76040262980911/1000000000
λ_{50}	$s(4,3,1)$	73824749/1000000000

Coercivity multipliers, 1–16

Entry		Exact rational value
μ_{1}	$s(*,0,0)$	71886295299/100000000
μ_{2}	$s(*,1,0)$	1908924932359/1000000000
μ_{3}	$s(*,2,0)$	1135980196793/500000000
μ_{4}	$s(*,3,0)$	1044723319169/500000000
μ_{5}	$s(1,*,0)$	1201394576459/1000000000
μ_{6}	$s(2,*,0)$	1292180672667/500000000
μ_{7}	$s(3,*,0)$	640272982733/200000000
μ_{8}	$s(1,0,0)$	8186883981967/50000000
μ_{9}	$s(1,2,0)$	279454961160469/1000000000
μ_{10}	$s(1,3,0)$	165328712036219/1000000000
μ_{11}	$s(2,0,0)$	140414654259099/1000000000
μ_{12}	$s(2,1,0)$	142895011420067/500000000
μ_{13}	$s(2,3,0)$	30720291963089/125000000
μ_{14}	$s(3,0,0)$	63498092323023/500000000
μ_{15}	$s(3,1,0)$	19233484226119/62500000
μ_{16}	$s(3,2,0)$	76017402005667/250000000

Coercivity multipliers, 17–32

Entry		Exact rational value
μ_{17}	$s(*,0,1)$	104536926291/125000000
μ_{18}	$s(*,1,1)$	2405242485183/1000000000
μ_{19}	$s(*,2,1)$	104340678109/50000000
μ_{20}	$s(*,3,1)$	165248337433/100000000
μ_{21}	$s(1,*,1)$	1969442541233/1000000000
μ_{22}	$s(2,*,1)$	2536896828057/1000000000
μ_{23}	$s(3,*,1)$	2473817086813/1000000000
μ_{24}	$s(1,0,1)$	103002747291613/500000000
μ_{25}	$s(1,2,1)$	208920702818967/1000000000
μ_{26}	$s(1,3,1)$	131376329777901/1000000000
μ_{27}	$s(2,0,1)$	34962082616757/250000000
μ_{28}	$s(2,1,1)$	91427294433371/500000000
μ_{29}	$s(2,3,1)$	7741389283089/40000000
μ_{30}	$s(3,0,1)$	141238249220977/1000000000
μ_{31}	$s(3,1,1)$	9147667423473/40000000
μ_{32}	$s(3,2,1)$	146712423316341/1000000000

B Exact positive-definiteness pivots

Let $A_D = T^\top(D - 10^{-6}Q)T$ and $A_C = C_{0:7,0:7}$. Starting from either A, repeatedly record $d = A_{00}$ and replace A by $A_{1:,1:} - A_{1:,0}A_{0,1:}/d$. The following are the exact eight pivots for each matrix in that order. Every numerator and denominator is a positive integer; line breaks inside an integer denote concatenation, not addition. This notation avoids illegibly scaled long fractions. Together with Appendix A and the reconstruction formulas, these are a finite exact positive-definiteness certificate.

Matrix A_D: pivots 1–4

Pivot 1.

Numerator:
`10287306338434339243`

Denominator:
`148537000000000`

Pivot 2.

Numerator:
`559407393367896517741892408855971`

Denominator:
`10287306338434339243000000000`

Pivot 3.

Numerator:
`71725839681682447489730269504565732962065620022705622 1`

Denominator:
`21271730173103934734291833019965277293312000000000`

Pivot 4.

Numerator:
`13569711108851435858602302717764341203700824157591282979782991250534881`

Denominator:
`183618149585107065573709489931688276382887987258126392576000000000`

Matrix A_D: pivots 5–8

Pivot 5.

Numerator:
```
1122630808789088955190854633184113944103120090030136868038547884572630531069902668
9
```

Denominator:
```
3487204461895269425829083458111698874349670100166325951501775383011591036500000000
```

Pivot 6.

Numerator:
```
2161785188367778882989731480580756168876846153816890894499536826472686698357721625922663494928
47
```

Denominator:
```
4152835887872597863042009459074674302026261837039482302248196334611074860533783952794880000000
00
```

Pivot 7.

Numerator:
```
5424385523240451393509818472328315840201407251556550285047000091850684764714512876293688586837736809821101
3
```

Denominator:
```
2568840692196678175541165959448190232451680777195996182370221612750187712911687177197397340369521187120000000
0
```

Pivot 8.

Numerator:
```
2968490650017040497057712733916075002052231394667803737749241224461974581968787327877400511608833264463372502549647817
```

Denominator:
```
1110914155159644445390810823132839084073248205118782469837762561881102023981353223706494742258436849865136154624000000
0
```

Matrix A_C: pivots 1–4

Pivot 1.

Numerator:
2225269966353277769

Denominator:
29707400000000

Pivot 2.

Numerator:
350648288137791982783988395926784 3

Denominator:
35604319461652444304000000000

Pivot 3.

Numerator:
351965220477384438931388903599535506329713612498351767

Denominator:
260421223875616038733926421828883797845500000000 0

Pivot 4.

Numerator:
780973340828661442088004181798136106617135971969815006398121627496889

Denominator:
2815721763819075511451111228796284050637708899986814136000000000 0

Matrix A_C: pivots 5–8

Pivot 5.

Numerator:
67416371558389864021067729894075481326367927094987404657576374385
706715683918781652476956 9

Denominator:
37121099880533403079496280688559917717948648277913731713714493498
72172844576000000000

Pivot 6.

Numerator:
47812007571786750026252401114605612024704937895160710420387781340
77751185141930179191669245154880422174631

Denominator:
43146477797369512973483347132208308048875473340791938980848879606
85229803770802025758525241600000000 0

Pivot 7.

Numerator:
27888596794407163164871661510526240687771056024525758326524386359
99805154246873690781005939786092532724228358327430810 1

Denominator:
44386576054315553054059080652251086208209983500834290273207124231
34418298671418018916206091672315457928457280293750000 0

Pivot 8.

Numerator:
17349682249592640944308940847075250925486027680187035836210927058
76284950863123673896527620440226390756757122912745969935003546 73

Denominator:
49579727634501623404216287129824427889370766265823570358265575751
10764718661108783610677226286386724843072637026543662400000000 0

C Reproduction and independent checking

The source archive contains the complete `exact_certificate.json`, the standalone `check_independent.py`, its exact result JSON, the manuscript, generated tables, table-generation script, style file, and build script. No numerical search output is required.

Run `python check_independent.py` in the extracted directory. Python 3 and SymPy are required; no optimization solver is used. The checker reconstructs interpolation labels independently, derives formal gradients from the literal update, verifies the 126-entry layout, both signed value identities, positivity of every multiplier, exact nullspace and support identities, the 16 strictly positive Schur pivots, all entries of M, its positive diagonals and positive determinant, and the exact threshold comparison. The expected terminal message begins `PASS`. Its result JSON prints the exact pivots and the witness SHA-256 digest.

The certificate's mathematical validity is the finite exact arithmetic described in Lemma 1; it does not depend on the method originally used to find its coefficients. The standalone checker independently reconstructs the certificate from the raw-map formulas and frozen witness. The analytic argument in Sections 2–7 is included in full, since a finite matrix check alone does not establish the analytic convergence conclusion.